

Save the Farm.

Hour 0 – 6 · Foundation Phase · Pass / Fail

You have 30 hours. Your mission is to design, build, and **defend** an edge-based IoT irrigation system the one that reads real sensors, controls real pumps, and makes real decisions. Not a prototype. Not a demo. A working system that survives whatever comes next.

Five challenge boxes are waiting. Each one introduces a new disaster. **You'll open them one at a time but only after a judge has cleared your foundation.**

Before anything else: build the base. Get it working. Prove it's real. Then call the judge.

Your Credentials as follows :

Raspberry Pi Login

Username : hgroup10

Password : hgroup10

Static IP : 192.168.200.20

****Please configure your Rasp Pi to this static IP**

SVN Repository

Repo URL : http://svn.home/svnrepo/hGroup10_Project/trunk

Username : hGroup10

Password : hgroup10svn

Minimum Database Requirement :

Table name: sensor_data

"id" — primary key, auto increment

"sensor_position" — varchar 255

"sensor_value" — decimal 10,2

"created_at" — timestamp

"If your sensors are hardcoded, your pump button does nothing, or you can't explain why your automation fires , you don't pass. There is no partial credit for a fake demo."

Build it right. Build it real. The clock is running.

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Your Objectives - ALL FOUR MUST PASS BEFORE OPENING BOX 1

- Sensor readings (min. 3 sensors, data polling every 1 minute)
- Dashboard exists
- Irrigation control (min. 1 pump)
- Basic automation logic (min. 1. example, EC low trigger water pump on)
- Min database requirement

Your Submission :

- Screenshot of dashboard (svn_url/docs/screenshot)
- Automation logic flowchart (svn_url/docs/flowchart)
- SVN commit -m "Foundation : Completed" --username [username] to central station

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PACKING LIST

20 items

No.	Photo	Item Name	Qty	Remark
1		ESP32 Smart Farm IoT Kit for Arduino	1	
2		Raspberry Pi 4 Model B - 2GB	1	
3		Small container x 4 compartment	1	
4		Adapter 12V 1A (UK Plug)	1	
5		Raspberry Pi 15W (5V/3A) PSU USB-C UK Plug - White	1	
6		Micro Submersible Water Pump DC 3V-5V	2	
7		2 Channel 5V Active Low Relay Module	1	
8		Breadboard 8.5 x 5.5 cm (400 Holes)	1	
9		40 Ways Male to Male Jumper Wire	1	
10		40 Ways Male to Female Jumper Wire	1	

No.	Photo	Item Name	Qty	Remark
11		40 Ways Female to Female Jumper Wire	1	
12		USB Type C Cable (1 Meter)	1	
13		MakerDisk Class 10 A1 microSD 32GB - Preloaded Raspberry Pi OS	1	
14		RS485/Modbus RTU 3 in 1 Soil Sensor - Moisture, Temperature & EC	1	
15		5V2A UK POWER SUPPLY ADAPTER CONVERTER	1	
16		DC Jack Female (2.1mm) to Screw Terminal Adapter	2	
17		USB to RS485 Converter	1	
18		LAN cable	1	
19		Push Button (Red) Panel Mount 10mm with Wire-20cm (SW-PBM-10R-W)	3	
20		Raspberry pi enclosure	1	